# cutting the lifi project

digitalisierung und programmierung · prof. dr. nicolas meseth

1 the elephant
2 cutting along the way

this is your
elephant.



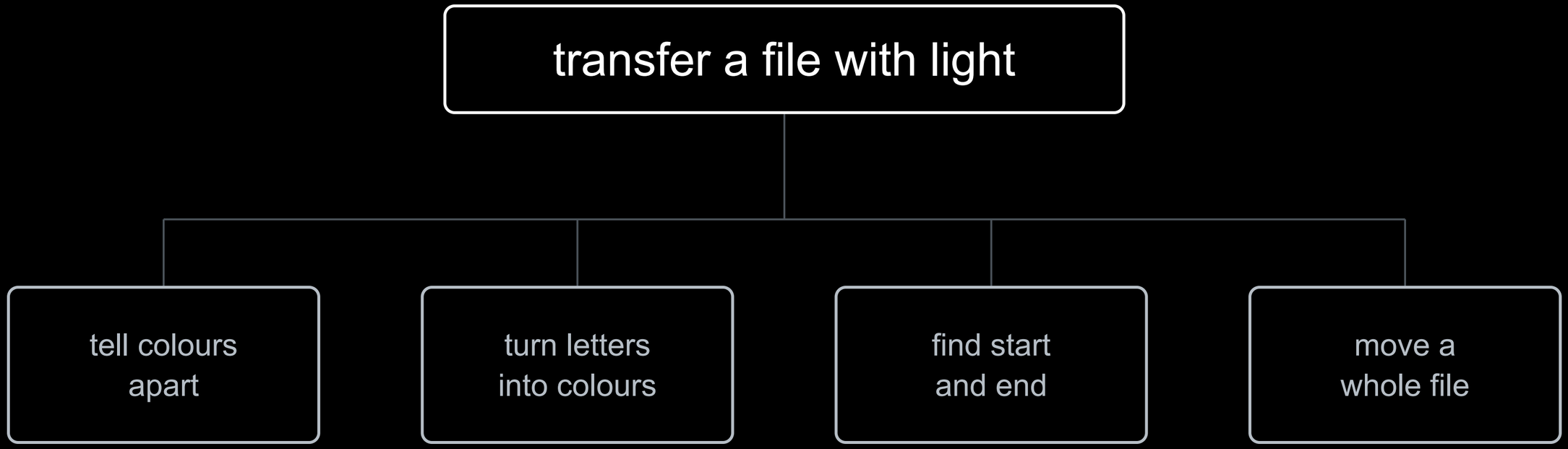
image: ai-generated (gpt-image-2.5-flare)

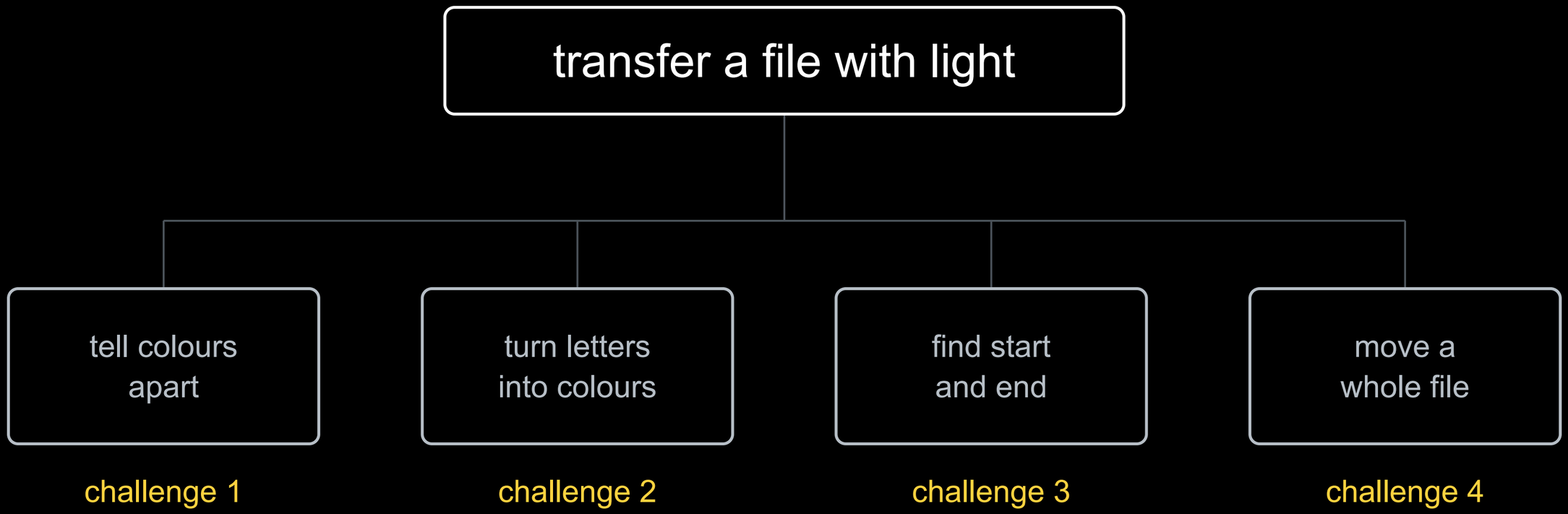
part 1

# the elephant

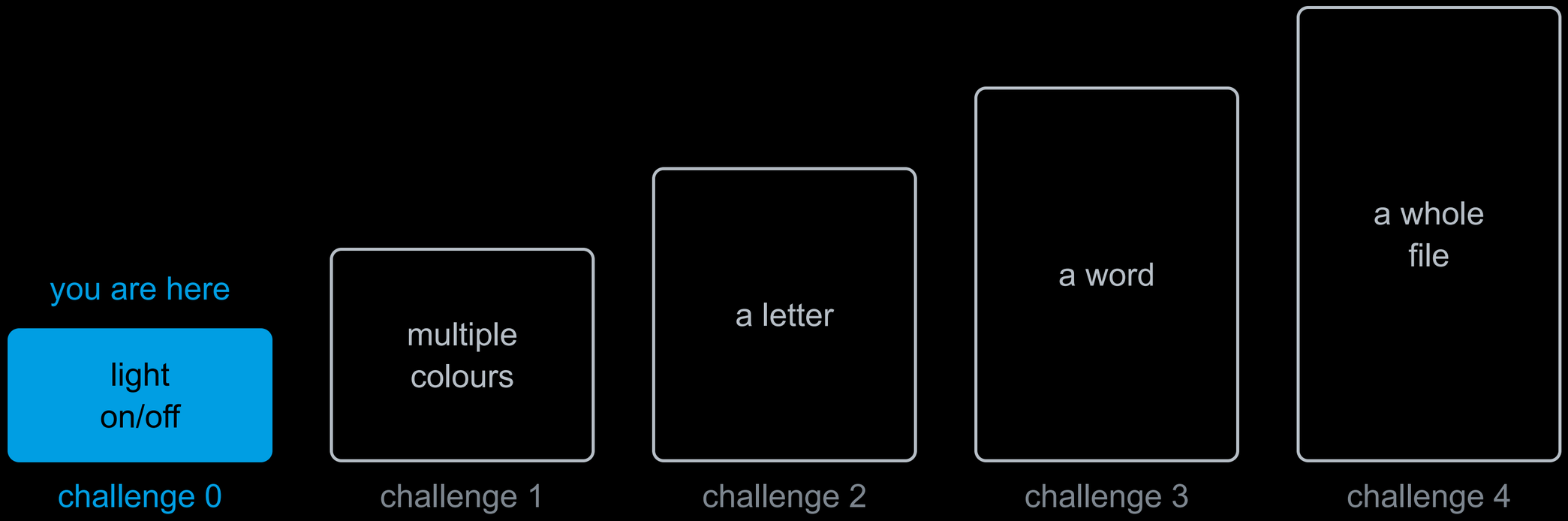
transfer a file from one laptop
to another, using light.

transfer a file with light


transfer a file with light
tell colours apart
turn letters into colours
find start and end
move a whole file


transfer a file with light
tell colours apart
turn letters into colours
find start and end
move a whole file
challenge 1
challenge 2
challenge 3
challenge 4

start smaller: the challenges are the stairs



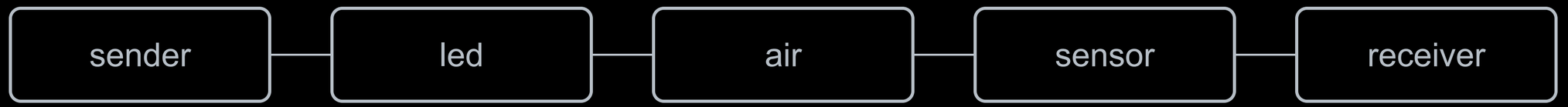
# how do computers solve complex problems?

- how do computers represent information?
  - colours into symbols, letters into bits
- how do computers store information?
  - a file is bytes; what holds them?
- how do computers transfer information?
  - signal and noise, timing, protocols
- how do computers process information?
  - checksums, filters, encryption

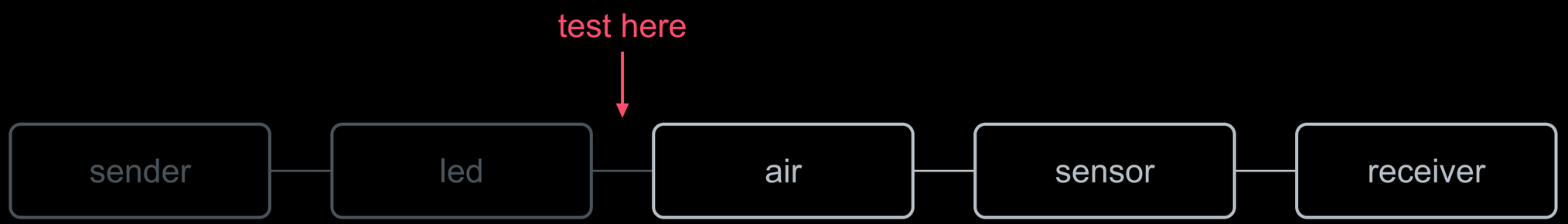
part 2

# cutting along the way

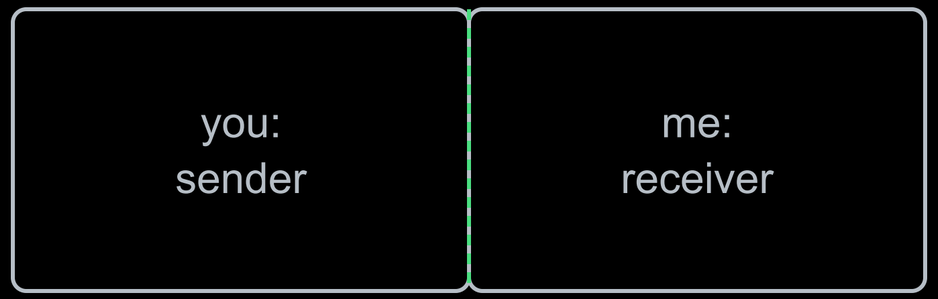
# when nothing arrives



# when nothing arrives



one person per end of the link




what crosses the border is written down

a testable piece of the project

the receiver works.

red or blue?
right 50 times in a row.

a testable piece of the project

the receiver works.

red or blue?
right 50 times in a row.



and the assistant gets small pieces

one big wish

```
build me the
file transfer.
```

one small piece, with its test

```
read the sensor ten times
and return the average.
test: covered, it stays under 10.
red led on, it goes over 100.
```

you cut the problem, the assistant solves the pieces, and you check each one.

and the assistant gets small pieces

one big wish

```
build me the
file transfer.
```

one small piece, with its test

```
read the sensor ten times
and return the average.
test: covered, it stays under 10.
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you cut the problem, the assistant solves the pieces, and you check each one.